Distribution of # degraded proteins

$$\frac{\partial G}{\partial t} = v_p \frac{\partial G}{\partial x} + k_d[A(x)](u_0 - G) = 0 \quad \text{at steady state}$$

(u_0: initial undeg prot)

$$G(L) = 0 \; ; \quad k_d[A(x)] = \frac{k_0 A(x)}{A(x) + K_M}$$

ss eqn:

$$\frac{dG}{dx} + \underbrace{\frac{k_0 A(x)}{v_p (A(x) + K_M)}}_{P(x)} (u_0 - G) = 0$$

$$\therefore \frac{dG}{dx} + P(x)(u_0 - G) = 0$$

multiply by $I = \exp(-\int P dx)$

$$I \frac{dG}{dx} + P(x) I(x) (u_0 - G) = 0$$

$$\Rightarrow \frac{d(I(u_0 - G))}{dx} = 0$$

$$\therefore \boxed{G(x) = u_0 - \frac{c}{I}}$$ (c: constant)

Boundary condition: $G(L) = 0$

$$I = \exp(-\int P dx) = \exp\left(-\int \frac{k_0 A(x)}{v_p (A(x) + K_M)} dx\right)$$

$$A(x) = C_0 \left(1 - \exp\left[\frac{\delta}{\beta}\left(e^{-\beta L} - e^{-\beta x}\right)\right]\right)$$